

Code No: R05010303

**R05****Set No. 2**

**I B.Tech Examinations, December-January, 2011-2012**  
**COMPUTER PROGRAMMING AND NUMERICAL METHODS**

Common to ME, CHEM, MECT, MEP, AME, MMT

Time: 3 hours

Max Marks: 80

**Answer any FIVE Questions**  
**All Questions carry equal marks**

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- Explain the different ways of passing structure as arguments in functions.
  - Write a C program to illustrate the method of sending an entire structure as a parameter to a function. [8+8]
- Find a real root of the equation  $x^3 - x - 4 = 0$  by bisection method.
  - Use iteration method to find a root of the equation  $x = \frac{1}{2} + \sin x$  [8+8]
- Temperature and pressure values are given from steam tables

|          |       |       |       |       |        |
|----------|-------|-------|-------|-------|--------|
| Temp °C  | 140   | 150   | 160   | 170   | 180    |
| Pressure | 3.685 | 4.854 | 6.302 | 8.076 | 10.225 |

Using Newton's interpolation formula find the pressure of the steam for a temp of 142° C

- Consider the following table for f(x)

|        |   |    |    |    |    |    |
|--------|---|----|----|----|----|----|
| x :    | 0 | 10 | 20 | 40 | 50 | 60 |
| f(x) : | 6 | 70 | 75 | 18 | 24 | 90 |

Calculate f(30) from the above table, using Lagrange and the divided difference Formula [8+8]

- Describe the two different forms of the if-else statement. How do they differ?
  - Compare the use of the if-else statement with use of the ?: operator. In particular, in what way can the ?: operator be used in place of an if-else statement?
  - Admission to a professional course is subject to the following conditions:
    - marks in maths  $\geq 60$ .
    - marks in physics  $\geq 50$ .
    - marks in chemistry  $\geq 40$ .
    - total in all three subjects  $\geq 200$ .

Given the marks in three subjects, write a C program to process the application to find whether eligible or not. [5+6+5]

- The annual examination is conducted for 50 students for three subjects. Write a program to read the data and determine the following:
  - Total marks obtained by each student.

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- (b) The highest marks in each subject and the Roll No. of the student who secured it.
- (c) The student who obtained the highest total marks. [5+6+5]
6. Use Runge-Kutta method to obtain an approximate solution to the differential equation  $\frac{dy}{dx} = y - x + 5$  at the points  $x=2.1, 2.2, 2.3$  with initial condition  $y(2)=1$ . [16]
7. (a) What is a pointer? How is a pointer initiated? Give an example.
- (b) State whether each of the following statements is true or false. Give reasons.
- i. An integer can be added to a pointer.
  - ii. A pointer can never be subtracted from another pointer.
  - iii. When an array is passed as an argument to a function, a pointer is passed.
  - iv. Pointers can not be used as formal parameters in headers to function definitions.
- (c) If  $m$  and  $n$  have been declared as integers and  $p1$  and  $p2$  as pointers to integers, then find out the errors, if any, in the following statements.
- i.  $p1 = \&m;$
  - ii.  $p2 = n;$
  - iii.  $m=p2-p1;$
  - iv.  $*p1 = \&n;$  [4+6+6]
8. Declare a queue of integers. Write functions
- (a) To insert an element in to queue
  - (b) To delete an element from queue [8+8]

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**R05****Set No. 4**

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Time: 3 hours

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1. The annual examination is conducted for 50 students for three subjects. Write a program to read the data and determine the following:

- (a) Total marks obtained by each student.
- (b) The highest marks in each subject and the Roll No. of the student who secured it.
- (c) The student who obtained the highest total marks. [5+6+5]

2. (a) Temperature and pressure values are given from steam tables

|          |       |       |       |       |        |
|----------|-------|-------|-------|-------|--------|
| Temp     | 140   | 150   | 160   | 170   | 180    |
| Pressure | 3.685 | 4.854 | 6.302 | 8.076 | 10.225 |

Using Newton's interpolation formula find the pressure of the steam for a temp of 142° C

- (b) Consider the following table for f(x)

|      |   |    |    |    |    |    |
|------|---|----|----|----|----|----|
| x    | 0 | 10 | 20 | 40 | 50 | 60 |
| f(x) | 6 | 70 | 75 | 18 | 24 | 90 |

Calculate f(30) from the above table, using Lagrange and the divided difference Formula [8+8]

3. (a) What is a pointer? How is a pointer initiated? Give an example.
- (b) State whether each of the following statements is true or false. Give reasons.
- i. An integer can be added to a pointer.
  - ii. A pointer can never be subtracted from another pointer.
  - iii. When an array is passed as an argument to a function, a pointer is passed.
  - iv. Pointers can not be used as formal parameters in headers to function definitions.
- (c) If m and n have been declared as integers and p1 and p2 as pointers to integers, then find out the errors, if any, in the following statements.
- i. p1 = &m;
  - ii. p2 = n;
  - iii. m=p2-p1;
  - iv. \*p1 = &n;
- [4+6+6]

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**R05****Set No. 4**

4. Use Runge-Kutta method to obtain an approximate solution to the differential equation  $\frac{dy}{dx} = y - x + 5$  at the points  $x=2.1, 2.2, 2.3$  with initial condition  $y(2)=1$ . [16]
5. (a) Explain the different ways of passing structure as arguments in functions.  
(b) Write a C program to illustrate the method of sending an entire structure as a parameter to a function. [8+8]
6. Declare a queue of integers. Write functions  
(a) To insert an element in to queue  
(b) To delete an element from queue [8+8]
7. (a) Find a real root of the equation  $x^3 - x - 4 = 0$  by bisection method.  
(b) Use iteration method to find a root of the equation  $x = \frac{1}{2} + \sin x$  [8+8]
8. (a) Describe the two different forms of the if-else statement. How do they differ?  
(b) Compare the use of the if-else statement with use of the  $?:$  operator. In particular, in what way can the  $?:$  operator be used in place of an if-else statement?  
(c) Admission to a professional course is subject to the following conditions:  
i. marks in maths  $\geq 60$ .  
ii. marks in physics  $\geq 50$ .  
iii. marks in chemistry  $\geq 40$ .  
iv. total in all three subjects  $\geq 200$ .

Given the marks in three subjects, write a C program to process the application to find whether eligible or not. [5+6+5]

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**R05****Set No. 1**

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1. The annual examination is conducted for 50 students for three subjects. Write a program to read the data and determine the following:
  - (a) Total marks obtained by each student.
  - (b) The highest marks in each subject and the Roll No. of the student who secured it.
  - (c) The student who obtained the highest total marks. [5+6+5]
2.
  - (a) What is a pointer? How is a pointer initiated? Give an example.
  - (b) State whether each of the following statements is true or false. Give reasons.
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  - (a) Describe the two different forms of the if-else statement. How do they differ?
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**R05****Set No. 1**

4. (a) Temperature and pressure values are given from steam tables

|                     |       |       |       |       |        |
|---------------------|-------|-------|-------|-------|--------|
| Temp <sup>o</sup> p | 140   | 150   | 160   | 170   | 180    |
| Pressure            | 3.685 | 4.854 | 6.302 | 8.076 | 10.225 |

Using Newton's interpolation formula find the pressure of the steam for a temp of 142<sup>o</sup> C

- (b) Consider the following table for f(x)

|        |   |    |    |    |    |    |
|--------|---|----|----|----|----|----|
| x :    | 0 | 10 | 20 | 40 | 50 | 60 |
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Calculate f(30) from the above table, using Lagrange and the divided difference Formula [8+8]

5. (a) Find a real root of the equation  $x^3 - x - 4 = 0$  by bisection method.  
 (b) Use iteration method to find a root of the equation  $x = \frac{1}{2} + \sin x$  [8+8]
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7. Use Runge-Kutta method to obtain an approximate solution to the differential equation  $\frac{dy}{dx} = y - x + 5$  at the points x=2.1, 2.2, 2.3 with initial condition y(2)=1. [16]
8. Declare a queue of integers. Write functions  
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Using Newton's interpolation formula find the pressure of the steam for a temp of 142° C

- (b) Consider the following table for f(x)

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| x    | : | 0 | 10 | 20 | 40 | 50 | 60 |
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5. Declare a queue of integers. Write functions
- (a) To insert an element in to queue
  - (b) To delete an element from queue [8+8]
6. Use Runge-Kutta method to obtain an approximate solution to the differential equation  $\frac{dy}{dx} = y - x + 5$  at the points  $x=2.1, 2.2, 2.3$  with initial condition  $y(2)=1$ . [16]
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